SAEON Product Backlog

Real Time Rainfall Monitoring System

The project follows 13 weekly sprints from 28 July to 23 October 2026, with work scheduled Tuesday to Friday. Planning, development, bench testing, documentation and presentation preparation are Completed. Final deployment, baseline field calibration and the 48-hour field validation activities remain In Progress.

Product Backlog	In Progress	Comple
Sprint 1: Project Initiation & System Architecture		
Day 1: Define Project Scope & Board Setup		Comple
Day 2: Component Verification & Sourcing		Comple
Day 3: Development Environment Configuration		Comple
Day 4: Architecture Diagram & Data Flow		Comple
Sprint 2: Core Firmware & Sensor Interfacing		
Day 1: ESP32 Base Setup & Serial Output		Comple
Day 2: DS18B20 Temperature Sensor Logic		Comple
Day 3: TR-525M Rain Gauge Logic		Comple
Day 4: Firmware Formatting & Low Power Mode		Comple
Sprint 3: HiveMQ Cloud Broker Setup		
Day 1: Broker Provisioning		Comple
Day 2: Topic Structuring & Access Control		Comple
Day 3: Network Connection & Publish Testing		Comple
Day 4: Database Storage Integration		Comple

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Sprints 4 to 6

Product Backlog	In Progress	Comple
Sprint 4: Cellular Communication (SIM7000G)		
Day 1: SIM Module Wiring & Power		Comple
Day 2: Network Bearer Configuration		Comple
Day 3: MQTT over Cellular		Comple
Day 4: Stability & Error Handling		Comple
Sprint 5: SAEON Dashboard Base Application		
Day 1: App Skeleton & Routing		Comple
Day 2: Real-Time Visualization		Comple
Day 3: Historical Data & Filtering		Comple
Day 4: Cloud Command Prep		Comple
Sprint 6: Power Supply & Solar Integration		
Day 1: Battery & Charge Controller Setup		Comple
Day 2: Solar Panel Integration		Comple
Day 3: Mains Power Alternative Testing		Comple
Day 4: System Power Flow Audit		Comple

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Sprints 7 to 9

Product Backlog	In Progress	Comple
Sprint 7: Advanced UI & Health Diagnostics		
Day 1: Develop Station Health Diagnostics UI Component		Comple
Day 2: Implement CSV Export Functionality		Comple
Day 3: Mobile & Tablet Responsiveness		Comple
Day 4: Full End-to-End Data Pipeline Ping Test		Comple
Sprint 8: Hardware Finalization & PCB Soldering		
Day 1: Circuit Transfer & Layout		Comple
Day 2: PCB Soldering		Comple
Day 3: Enclosure Preparation		Comple
Day 4: Component Mounting & Sealing		Comple
Sprint 9: Pre-Deployment Auditing & Calibration		
Day 1: Power Drain Audit		Comple
Day 2: Solar Charging Verification		Comple
Day 3: Full System Deployment	In Progress	
Day 4: Baseline Calibration	In Progress	

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Sprints 10 to 12

Product Backlog	In Progress	Comple
Sprint 10: 48-Hour Field Stress Test		
Day 1: Stress Test Initiation	In Progress	
Day 2: Active Monitoring	In Progress	
Day 3: Manual Field Measurements	In Progress	
Day 4: Data Auditing & Conclusion	In Progress	
Sprint 11: Presentation Formatting & Content		
Day 1: Presentation Outlining		Comple
Day 2: Technical Diagrams & Architecture Slides		Comple
Day 3: Dashboard & Software Slides		Comple
Day 4: Technical Documentation		Comple
Sprint 12: Rehearsal & Submission Prep		
Day 1: Slide Deck Finalization		Comple
Day 2: Group Rehearsal 1		Comple
Day 3: Timed Rehearsal & Feedback		Comple
Day 4: Final Compilation		Comple

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Sprint 13

Product Backlog	In Progress	Comple
Sprint 13: Project Buffer & Final Review		
Day 1-4: Buffer & Prototype Documentation Sign-off		Comple

Status and source notes

The statuses below reflect the project owner's updates. Trello is the task-tracking board; GitHub stores code, documentation and evidence. The sprint backlog uses the same 13-sprint schedule and task statuses. These document updates do not modify the live Trello board.

Workflow: Product Backlog, In Progress and Completed. Sprint headings organise tasks and do not have completion marks. Sprint 13 sign-off covers the prototype documentation and presentation package; it is not final field-deployment acceptance. Supabase, historical data, CSV export, battery monitoring and solar integration are included in the current project scope.

Archived lists and archived cards are excluded. This product backlog shows sprint and daily card titles; individual checklist subtasks remain in Trello.

Source: SAEON Real-Time Rainfall Monitoring System Trello board export supplied by Ethan Williams.